

Review

A systematic review of anal squamous cell carcinoma in inflammatory bowel disease

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ABSTRACT

Aim: The aim of this systematic review was to determine the incidence, aetiology and clinical characteristics of anal squamous cell carcinomas (SCC) presenting in patients with inflammatory bowel disease. **Method:** A systematic review of the literature was undertaken using Medline, Embase, Cochrane and Web of Science.

Results: A total of 33 cases of anal SCC were described, 7 in ulcerative colitis (UC) and 26 in Crohn's disease (CD). The annual incidence of anal SCCs was 0.9/100,000 and 2.0/100,000 in patients with UC and CD respectively. The gender ratio in CD was 3M:17F with a median age of 42 years, the main presenting symptom was anal pain and 85% of CD cases had peri-anal disease. No studies described anal intra-epithelial neoplasia. The human papilloma virus was found to be positive in 2 out of 5 (40%) cases. The majority of patients (73%) with CD received radical surgery as their first line treatment. The cumulative overall and disease free survival in CD was 37 per cent at five years.

Conclusion: The findings of this review when contrasted with the data from cancer registries suggests that there is a higher incidence of anal SCC, an earlier age of presentation and poorer outcomes in patients with Crohn's disease compared to the general population implying a more aggressive neoplastic process. This review supports the hypothesis that peri-anal disease plays a contributing role in anal SCCs and as such targeted surveillance in patients with longstanding peri-anal disease should be considered.

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Introduction

The majority of anal carcinomas are squamous in origin and have an approximate annual incidence of 1 in 100,000 cases per year in the heterosexual population [1,2]. The exact incidence of anal SCCs in patients with inflammatory bowel disease (IBD) is currently unknown due to a paucity of reported cases [3,4]. There have been conflicting reports of the incidence of anal SCCs in the literature with some studies demonstrating a high incidence of anal SCC in IBD patients whereas other studies have demonstrated no appreciable difference [5,6]. The precise aetiology underpinning the development of anal SCCs is still not fully understood but known risk factors for the development of anal carcinoma include: smoking, sexual behaviour, human papilloma virus (HPV), immunosuppression and the presence of anal condylomata acuminata [2,7–9]. It is widely accepted, however, that HPV plays a key role in the development of both anal intra-epithelial neoplasia (AIN) and anal squamous cell carcinomas particularly in immuno-deficient patient groups [2]. Patients with inflammatory bowel disease are commonly immunosuppressed as a result of both the underlying disease process as well as the common treatment regimes they undergo. The aim of this review was to determine the incidence, aetiology and clinical characteristics of anal squamous cell carcinomas in patients with inflammatory bowel disease.

Method

A systematic review of the literature was performed to assess: (1) the incidence of AIN in patients with inflammatory bowel disease; (2) the incidence of anal squamous cell carcinomas in patients with inflammatory bowel disease; (3) the incidence of HPV associated AIN or anal squamous cell carcinomas in patients with inflammatory bowel disease; (4) to describe the demographic and clinical characteristics of inflammatory bowel disease sufferers presenting with anal squamous cell carcinomas. The methodology undertaken was based on the guidelines from the Preferred Reporting Items for systematic Reviews and Meta-Analyses (PRISMA) statement [10].

Search strategy

An electronic database search was performed in February 2013 using the following resources: MEDLINE from PubMed (1950 to present), Embase, Classic Embase, Cochrane and the Web of Science. The following search headings were used: “inflammatory bowel disease”, “ulcerative colitis”, “Crohn’s” combined with the Boolean operator ‘AND’ and each of the following terms: “squamous cell carcinomas”, “human papilloma virus” and “anal intra-epithelial neoplasia”. The titles were initially scanned and abstracts of interest were reviewed. All articles reviewed and included in the study had their reference lists scanned and studies found were included in the study if they met the inclusion criteria. No language restrictions were used. Foreign languages were translated using Google translate or by the authors if known.

Definitions

It was previously common practice to differentiate histologically between distal and proximal anal canal squamous cell carcinomas [1]. In the literature, previous nomenclature used to describe proximal anal canal squamous cell carcinomas included the terms “basaloid” or “cloacogenic tumours” [1]. For the purpose of this review anal squamous cell carcinomas were considered to be all anal canal squamous cell carcinomas including tumours described as “basaloid” or “cloacogenic”.

Inclusion criteria

To enter the review the study had to: 1) include histologically proven anal intra-epithelial neoplasia or anal squamous cell carcinomas in patients in adult patients with inflammatory bowel disease; 2) describe either the incidence or the clinical characteristics of anal squamous cell carcinomas in inflammatory bowel disease.

Exclusion criteria

The following criteria were used to exclude studies from the review: 1) studies that described anal cancers of non-squamous origin; 2) where the squamous cell carcinoma arose in a fistula or outside of the anal canal; 3) where there was significant overlap in the cohorts of patients from the same institutions or duplicate studies; 4) articles that were conference abstracts, editorials/commentaries or reviews.

Data extraction

Two authors (AS, AB) independently collected and tabulated the data into an electronic spreadsheet. Any differences in collated data between the two authors were discussed and agreement was reached by consensus. The specific data items collected were the following: first author/institute, year of publication, year of study, country of origin, study design, period of study, study sample size, proportion of patients with anal squamous cell carcinoma, duration of the inflammatory bowel disease (<10 yrs or ≥ 10 yrs), use of immunosuppressants/steroids for inflammatory disease, duration of medical therapy (<5 yrs or ≥5 yrs), presenting symptoms, method of diagnosis, use of neo-adjuvant/adjuvant therapy, use and type of surgery, post-operative histology, presence of anal intra-epithelial neoplasia or human papilloma virus, overall survival and recurrence.

Data analysis/quality of studies

Studies that met the inclusion criteria were assessed for quality and graded for their level of evidence according to the recent grading system of the Scottish Intercollegiate Guidelines Network (SIGN) [11]. It was elected to perform a descriptive review of the data as opposed to a meta-analysis due to the limited number of studies, heterogeneity of the populations described and the lack of comparative studies.

Statistical analysis

Statistical analysis was undertaken using SPSS version 19 (SPSS, Chicago, Illinois, USA). Due to the limited number of reported cases no comparative statistical analysis could be performed. Descriptive analyses were used to generate means and medians for continuous variables. Crude annual incidence was calculated using only studies where cohorts of greater than one hundred patients with IBD had been included. The following calculation was used: Annual incidence = ((Total number of anal SCC cases in IBD/total number of IBD cases))/((Cumulative length of cohort follow up/number of studies)). Survival tables were used to assess the cumulative proportions surviving at 3 and 5 years for both overall survival and disease free survival.

Results

Study selection, design and quality

A total of 1107 citations including references were found. After exclusion of duplicate citations, a review of the titles and abstracts resulted in 53 articles being reviewed of which 11 met the inclusion criteria (Fig. 1). One epidemiological study and three case series had to be excluded due to an inability to extract data in one study (12), overlap of a cohort in another (6), and a duplication of cases in two

studies [12,13]. All of the studies were retrospective in nature, eight were case series or case reports (evidence level 3) [4,5,14–20], one was a cohort study (evidence level 2++) [21] and two were case control studies (evidence level 2–/2+) [5,22] published between 1980 and 2010 (Table 1). The quality scored as acceptable for the two case control studies whilst the cohort study was scored as high quality (Table 1). Six of the studies had sample sizes greater than one hundred patients with inflammatory bowel disease [4,5,14,19,21,22]. Nine of the studies were from either the USA or European countries whilst there was one study from Australia and South Korea respectively (Table 1).

Demographics

A total of 33 cases of anal squamous cell carcinoma were described in patients with inflammatory bowel disease (Table 1). One study described seven cases of anal SCC in patients with ulcerative colitis but provided no demographic or clinical data [21]. The remaining 26 cases of anal squamous cell carcinomas were described in patients with Crohn's disease (Table 1). The rest of the results are described for the cases of anal SCC in Crohn's patients only. The gender ratio was 3M:17F with a median age of 42.00 (27–80) ± 14.5. The median age was 79.0 (41–80) and 41.0 (27–65) for males and females respectively. The annual incidence of anal SCC was calculated as 0.002% per annum which equates to 2.00 cases a

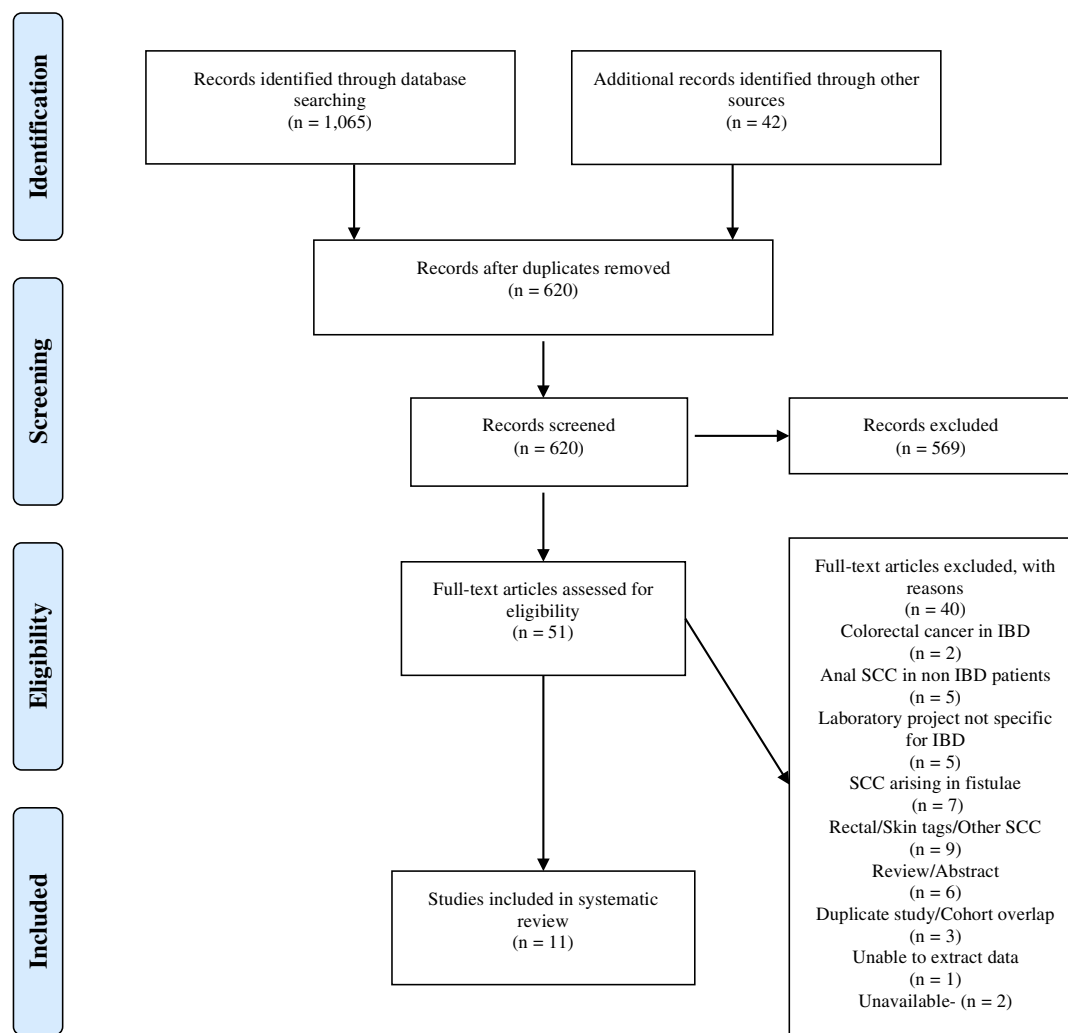


Figure 1. Modified PRISMA flow diagram demonstrating results of systematic literature search.

Table 1

Study characteristics and patient demographics of reported cases of anal squamous cell carcinomas in inflammatory bowel disease.

Author	Year	Country	Study	Period of study	Sample size	SCC		Gender		Age	Age		SIGN	§
						CD	UC	M	F		M	F		
Kang J et al. [4]	2010	S. Korea	Case report	1991–2008	843	1	n/a	0	1	43*	—	43*	3	n/a
Connell WR et al. [14]	1994	UK	Case series	1940–1993	2480	5	n/a	2	3	53	60.5	53	3	n/a
Church JM et al. [15]	1985	USA	Case series	1980–1985	4	1	n/a	0	1	65*	—	65*	3	n/a
Devon KM et al et al [16]	2009	USA	Case series	—	14	3	n/a	0	3	46	—	46	3	n/a
Slater G et al. [5]	1983	USA	Case control	1960–1981	1227	3	n/a	0	3	40	—	40	2–	+
García Díaz E et al. [17]	2000	Spain	Case report	—	1	1	n/a	0	1	46*	—	46*	3	n/a
Daly JJ et al. [18]	1980	USA	Case report	—	1	1	n/a	0	1	36*	—	36*	3	n/a
Lumley JW et al. [19]	1991	Australia	Case report	1985–1990	127	1	n/a	0	1	33*	—	33*	3	n/a
Walgenbach S et al. [20]	1987	Germany	Case series	—	2	2	n/a	0	2	34	—	34	3	n/a
Sunesen, K G et al. [21]	2010	Denmark	Cohort study	1978–2005	40,535 (CD-12609 UC- 27,926)	6	7	—	—	—	—	—	2++	++
Connell WR et al. [22]	1994	UK	Case control	1962–1991	755 (CD-450 UC-282)	2	0	1	1	59.5	79*	40*	2+	+

* Denotes single values all other values are means; M- male; F- female. §- quality score: (+)- acceptable quality; (++)-high quality; CD- Crohn's disease; UC- ulcerative colitis; SCC- squamous cell carcinoma.

year per 100,000 patients with Crohn's disease. The annual incidence of anal SCC in ulcerative colitis was calculated as 0.0009 per annum equating to 0.90 cases a year per 100,000 ulcerative colitis patients.

Inflammatory bowel disease

The duration of the Crohn's disease was described in nine studies (20 cases) and was greater than ten years in 16 (80.0%) cases (Fig. 2). The presence of perianal disease/fistulae was described in ten studies (20 cases), 17 (85.0%) patients had either peri-anal disease or fistulae of which the duration was greater than ten years in 11 (64.7%) patients (Fig. 2). Medical treatment was described in four studies (seven cases), six (85.7%) patients received steroids/immuno-suppressants and the duration of use was greater than five years for all six patients (Table 2).

Anal cancer in Crohn's disease

The presenting symptoms were described in 8 studies (18 cases), the most frequent symptom was anal pain in 12 (66.7%) cases (Fig. 3) [4,5,14–18,20]. The method of diagnosis was described in eight studies (15 cases) and was pre-operative in 11 (73.3%)

cases, post-operative in 3 (20.0%) cases and intra-operative in 1 (6.7%) case (Table 2). No cases of anal intra-epithelial neoplasia were described in any of the studies. Human papilloma virus was tested for in two studies (five cases), two (40%) patients tested positive for HPV (Table 2). The differentiation of the tumours was described in six studies (10 cases), seven (70%) were well differentiated, one (10%) was moderately differentiated and two (20%) were poorly differentiated (Table 2). The resection margins and the tumour stage were described in one study (Table 2). The use of a surgical intervention was described in nine studies (18 cases) of which three (16.7%) received no surgery (Table 2). In the patients receiving surgery, 11 (73.3%) received radical surgery (Fig. 4). The use of adjuvant or neo-adjuvant therapy was described in nine studies (18 cases), six (33.0%) patients received either neo-adjuvant therapy or adjuvant therapy (Table 2).

Survival

Survival was described in six studies (17 cases), median overall survival was 5.0 years and the cumulative proportion of overall survival at three and five years was 43% and 37% respectively (Table 2). The rate of recurrence was described in five studies (11 cases), the median disease free survival was 3.96 years and the

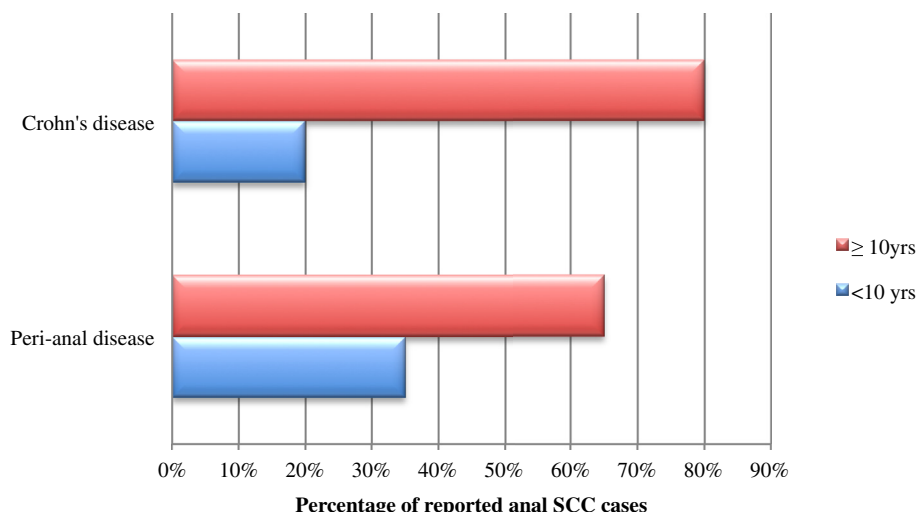


Figure 2. Bar chart demonstrating the duration of Crohn's disease and peri-anal disease in reported cases of anal squamous cell carcinomas.

Table 2
Clinical characteristics of patients presenting with anal squamous cell carcinomas in inflammatory bowel disease.

Author	SCC	CD ≥ 10 yrs	Peri-anal disease ≥ 10 yrs	Peri-anal HPV + ve disease	Diagnosis	Steroids/immunosuppressants	Therapy ≥ 5yrs	Treatment	Surgery	Histology	Stage	RO Survival	Recurrence
Kang J et al. [4]	1	Yes	Yes	–	Pre-op	Yes	Yes	Neo-adjuvant	No	G2	–	n/a	–
Connell WR et al. [14]	5	Yes- all	Yes- all	0/2	Pre-op-4	2	Yes	Neo-adjuvant-1	TPC-1 Local-2 Colostomy-1	G3-2	–	–	Alive-3 (0.5/3/7 yrs) 0 Dec-2 (0.7/11 yrs)
Church JM et al. [15]	1	Yes	Yes	–	Pre-op	–	–	Neo-adjuvant	No	–	–	n/a	Dec- 1.6 yrs
Devon KM et al. [16]	3	Yes- all 2 (67%)	Yes	2/3	Pre-op-1 Post-op-2	–	–	Neo-adjuvant-1	TPC-2 APR-1	G3- 2, G1- 1	II- 2 IV-1	1/3	Alive-2 (2.2/3.1 yrs) 2 Dec-1 (1.83 yrs)
Slater G et al. [5]	3	1	Yes- all	1	–	–	–	Neo-adjuvant-1	Local-1 APR-2	–	–	–	2 (1.0/2.5 yrs)
Garcia Diaz E et al. [17]	1	Yes	Yes	–	Pre-op	–	–	Adjuvant-1	TPC	G3, N0	M0	–	–
Daly JJ et al. [18]	1	No	1	–	Intra-op	–	–	Neo-adjuvant	TPC	In situ	–	–	–
Lumley JW et al. [19]	1	Yes	1	–	Pre-op	Yes	Yes	Adjuvant	TPC	G1, N0	M0	–	0.75 yrs
Walgenbach S et al. [20]	2	1	Yes- all	1	Pre-op-2	–	–	Adjuvant-2	APR-2	G3- 2	–	–	2 (0.25/0.3 yrs)
Sunesen, K G et al. [21]	13 ^a	–	–	–	–	–	–	–	–	–	–	–	–
Connell WR et al. [22]	2	Yes- all	Yes- all	–	–	Yes- all	Yes	–	–	–	–	–	–
Totals (% of reported cases)	33	16 (80)	19 (95)	12 (67)	Pre-op-11 (73.3) Intra-op-1 (6.7) Post-op-3 (20)	6 (67)	6 (100)	Neo-adjuvant-6 (33) Adjuvant-6 (33)	TPC/APR-11 (73%)	G1-2 (20) G2-1 (10) G3-7(70)	n/a	n/a	Recurrence- 5 (45.5)

SCC- squamous cell carcinoma; CD- Crohn's disease; HPV- human papilloma virus; TPC- total proctocolectomy; local- local excision; G1- well differentiated; G2- moderately differentiated; G3- poorly differentiated.

^a Denotes only study to describe cases in both ulcerative colitis and Crohn's disease, all other studies describe cases in Crohn's disease only.

cumulative proportion of disease free survival was 40% and 37% at three and five years respectively (Table 2).

Discussion

The overall level of the evidence available in the literature was poor with few reported cases of anal squamous cell carcinomas limiting both the interpretation and scope of the review. This review, however, serves to demonstrate that anal squamous cell carcinomas are rare in inflammatory bowel disease. In particular, this is the case in patients with ulcerative colitis where the annual incidence rate in ulcerative colitis patients was found to be similar to the general population. This is consistent with two studies undertaken by Frisch et al. whereby the incidence of anal SCCs in ulcerative colitis was found to be only slightly elevated compared to the general population [6,23]. The annual incidence of anal cancer in the general population from the Surveillance, Epidemiology, and End Results (SEER) data between 1975 and 2009 was 1.2 per 100,000 [24]. In contrast the incidence of 2 per 100,000 found in this review suggests that the annual incidence rate in Crohn's disease is elevated compared to the general population. This is consistent with the literature, as although there have been conflicting reports [6,23], the majority of studies have demonstrated an increased incidence of anal SCCs in Crohn's disease [5,12,21,22]. The median age of patients with Crohn's disease presenting with anal SCC in this review was 42 whereas the median age of presentation of anal cancer in the general population from the Danish Cancer Registry (1978–2008) was 65 [9]. Furthermore, recent data from SEER demonstrates that the highest incidence of anal cancer was found in the 50 to 64 and the ≥65 age groups in males and females respectively [25]. These findings would suggest that anal cancer may present earlier in patients with Crohn's disease. It is generally accepted that in the Crohn's population the presentation of anal squamous cell cancers tends to occur at an earlier age than in the general population [16]. Interestingly, we found that the age of presentation was earlier in females compared to male patients. This trend was not demonstrated in the data from the Danish Cancer Registry (1978–2008) [9]. This finding may be as a result of the small number of reported cases of anal SCC in males as opposed to a biological reason. In the general population, anal SCCs tend to be seen more frequently in females than males [1,26]. As such it was not unexpected to find a higher female to male ratio in the Crohn's patients with anal SCCs.

The human papilloma virus infects keratinocytes secondary to damage at the squamo-columnar junction and is considered to be the most important cause of anal squamous cell carcinoma [1,27–30]. The fact that numerous different variants of the HPV virus exist has been known since the seventies, it is now currently accepted that at least 130 different low risk and high risk variants exist [31]. High risk variants, which are capable of integrating into the host genome, include HPV 16 and 18 which are commonly associated with anal squamous cell carcinomas [6,32,33]. Integration of the oncogenes E6 and E7 into the host genome is the initial crucial step leading to carcinogenesis, E6 induces degradation of p53 whilst E7 inactivates products of the retinoblastoma family [27,34,35]. It is thought that the HPV infection is the initiating step in anal carcinogenesis leading to anal intra-epithelial neoplasia with further stepwise genetic alterations required to develop invasive carcinoma via either the chromosomal instability or the microsatellite instability pathways [8]. It is known that immunosuppression as well as high risk variants of HPV are risk factors for the progression of AIN and the development of anal SCC [30,36,37]. Interestingly, a PCR based study demonstrated that all the anal SCCs of patients who were immunosuppressed as a result of treatment for IBD were found to be positive for high risk HPV variants [30].

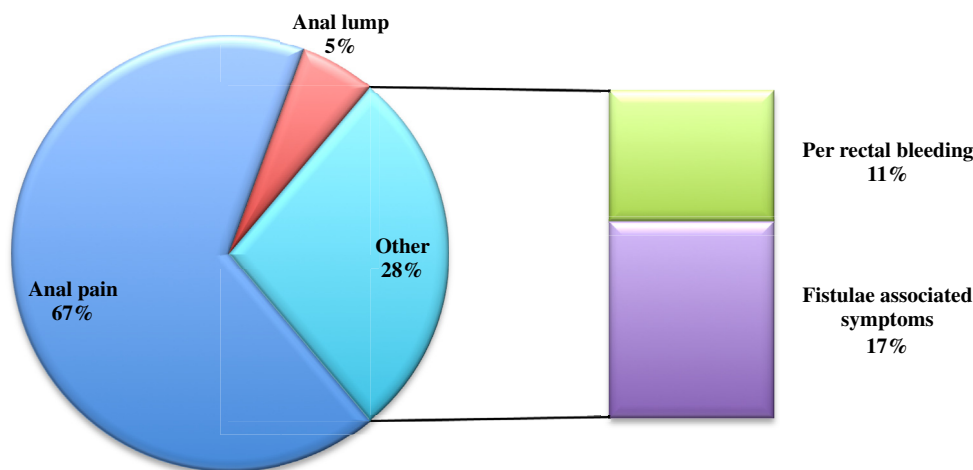


Figure 3. Pie chart demonstrating presenting symptoms of anal squamous cell carcinomas presenting in Crohn's disease.

The use of immunosuppressive therapy is becoming more common in patients with IBD and we thus hypothesised that there would be a high rate of HPV associated AIN and invasive carcinomas [38]. Regrettably, none of the studies reviewed described the incidence of AIN in inflammatory bowel disease whilst only two studies specifically tested for the presence of HPV making a causative link in Crohn's disease difficult to establish. Since 2006, there has been a prophylactic quadrivalent vaccine against HPV types 6,11,16 and 18 using a non-live vaccine. [29] The consensus statement from the European Crohn's and Colitis Organisation only recommends vaccination in females suffering from IBD to prevent cervical cancer [29]. There is a similar situation in the USA where there are no specific recommendations for HPV prophylaxis for anal SCC in IBD sufferers [39].

It is postulated that the presence of peri-anal disease in Crohn's disease may pre-dispose to the development of anal SCCs. It is thought that the chronic inflammation related to the peri-anal fistulae results in a dysplasia-carcinoma sequence [5,15,40]. Indeed there have been several reports of SCCs arising in peri-anal fistulae [41]. More recently an alternative hypothesis is that the epithelialized surfaces of the peri-anal fistulae may provide an entry point for the HPV virus [16,42]. Several studies provide credence to this theory demonstrating that patients with peri-anal disease have a greater risk of developing anal cancer and high risk variant HPV related anal SCCs [6,43]. The presence of peri-anal disease was found in the majority of the reported cases supporting a possible pre-

disposing or contributing role. The presence of peri-anal disease in Crohn's disease is common and may explain why the incidence is higher in Crohn's disease than in ulcerative colitis despite both sets of patients being commonly immuno-suppressed. Since the early seventies, it has been known that a prolonged duration of Crohn's disease increases the risk of developing carcinoma [44]. This is consistent with the findings of this review where the duration of both the Crohn's disease and the peri-anal disease were found to be greater than ten years in the majority of cases.

The diagnosis of anal SCCs can be challenging in Crohn's disease patients particularly as the presence of peri-anal disease can masquerade the presenting symptoms [16,45]. This was evident from the reported cases of anal SCC in Crohn's disease where the main presenting symptom was anal pain. In view of this, new or deteriorating symptoms despite medical therapy should lead to a high index of suspicion for anal SCC and should lead to appropriate radiological imaging and a prompt EUA [4,16]. Abdomino-perineal resections were the traditional first line of treatment for anal SCCs. However, it is now widely accepted that chemo-radiation should be the initial standard of care for patients with anal carcinoma [26,46]. Local excision can be undertaken for T1N0 (<2 cm) well differentiated tumours [26]. Abdomino-perineal resections tend to be only performed in cases where radiotherapy is contraindicated or as a salvage procedure for recurrences or progressing primaries despite oncological treatment [26,47]. The main prognostic indicator in these cases is to achieve an R0 resection [47]. The

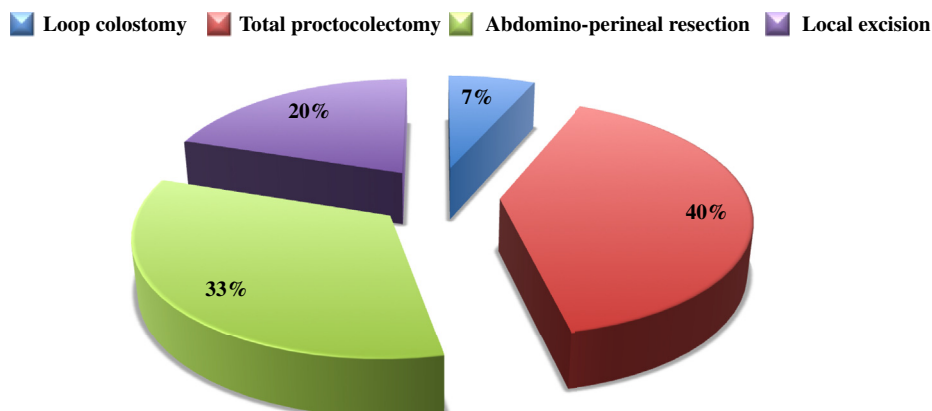


Figure 4. Pie chart demonstrating surgery performed for anal squamous cell carcinomas in Crohn's disease.

reporting of the post-operative histology, tumour stage and the resection margins in the studies reviewed was very poor limiting our interpretation of the reported findings. In patients with Crohn's disease, radiotherapy is commonly precluded due to underlying sepsis whilst the primary tumours tend to be advanced and generally require radical surgery [16,48]. This may help to explain the poor five year overall survival and disease free rates seen when compared to the overall survival and disease free survival of between 65 and 75% and 60% that can be achieved at five years with modern chemo-radiotherapy regimes [49]. Indeed, the five year survival of 37 per cent found in this review is much lower in contrast to the five year survival rate of 65.4 per cent described in the SEER data (1988–2008) possibly suggesting that patients with Crohn's disease have poorer outcomes than the general population [50]. Despite the poorer outcomes seen in Crohn's disease patients the expert consensus from the American Gastroenterological Association recommends that 'standard oncologic surgical principles should be followed' whilst there are no specific recommendations for patients with Crohn's disease in the European Society for Medical Oncology statement on anal cancer [3,26].

Conclusion

This systematic review demonstrates that anal squamous cell carcinoma in inflammatory bowel disease is a rare entity. Its incidence appears to be raised in patients with Crohn's disease whereas there does not appear to be a demonstrable difference between ulcerative colitis patients and the general population. The findings from this review when contrasted with the data from cancer registries suggest that anal squamous cell carcinomas in Crohn's disease present at an earlier age and are associated with poorer outcomes compared to the general population implying a more aggressive neoplastic process. Our understanding of the underlying aetiology of anal squamous cell carcinomas in Crohn's disease and in particular the role of the human papilloma virus remains limited due to the rarity of the condition. The majority of reported cases, however, had peri-anal disease of long duration supporting the hypothesis that this may contribute to the development of anal squamous cell carcinomas. In the absence of comprehensive guidelines for this group of patients, targeted surveillance of patients with longstanding peri-anal disease should be considered.

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This work received no funding.

Authorship statement

Guarantor of the integrity of the study: A. Slesser.

Study concepts: P. Tekkis and A. Slesser.

Study design: P. Tekkis, A. Slesser, A. Bhangu, R. Goldin and M. Bower.

Definition of intellectual content: A. Slesser, A. Bhangu, M. Bower, R. Goldin and P. Tekkis.

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Manuscript review: A. Slesser, A. Bhangu, M. Bower, R. Goldin and P. Tekkis.

Conflict of interest statement

The authors have none to declare.

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